Whole-Space Nonattainment and Strict Spreading for a Positive Curvature Functional

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Abstract

For a nonnegative energy comprising a Dirichlet term and a bounded, nonlinear weight multiplying the squared Laplacian, we show that the fixed-positive-mass infimum on $H^2(\mathbb{R}^d;\mathbb{C})$ is zero and is not attained in any finite spatial dimension $d \geq 1$. For $1 \leq d \leq 4$, the energy is strictly increasing along the mass-preserving dilation parameter, so no positive-mass local constrained minimizer exists in the strong H^2 topology. The conclusions are variational and do not exclude stationary saddle points or localization supplied by other terms or boundary conditions.

1 Functional and results

Fix $d \in \mathbb{N}$ and constants $c_1, c_2, \varepsilon > 0$ and $\alpha \geq 0$. For $s \geq 0$ define

$$F(s) = \frac{s}{\left(s + \varepsilon^2 e^{-2\alpha s}\right)^2}. \tag{1}$$

For $\psi \in H^2(\mathbb{R}^d;\mathbb{C})$, set

$$E(\psi) = c_1 \int_{\mathbb{R}^d} |\nabla\psi|^2 \,\mathrm{d}x + c_2 \int_{\mathbb{R}^d} F(|\psi|^2)|\Delta\psi|^2 \,\mathrm{d}x. \tag{2}$$

For every mass $m > 0$ define

$$\mathcal{A}_m = \{\psi \in H^2(\mathbb{R}^d;\mathbb{C}) : \|\psi\|_{L^2}^2 = m\}, \qquad I_d(m) = \inf_{\psi\in\mathcal{A}_m} E(\psi). \tag{3}$$

Theorem 1 (Positive-mass whole-space nonattainment)**.** *For every finite dimension $d \geq 1$ and every $m > 0$, the energy (2) is finite and nonnegative on $H^2(\mathbb{R}^d;\mathbb{C})$, strictly positive for each nonzero $\psi \in H^2(\mathbb{R}^d;\mathbb{C})$, and*

$$\boxed{I_d(m) = 0, \qquad \arg\min_{\psi\in\mathcal{A}_m} E(\psi) = \varnothing.} \tag{4}$$

Theorem 2 (Strict spreading for dimensions one through four)**.** *Suppose $1 \leq d \leq 4$. For each nonzero $\psi \in H^2(\mathbb{R}^d;\mathbb{C})$ and $\lambda > 0$, let*

$$\psi_\lambda(x) = \lambda^{d/2}\psi(\lambda x). \tag{5}$$

The function $\lambda \mapsto E(\psi_\lambda)$ is continuously differentiable and strictly increasing on $(0,\infty)$. In particular,

$$0 < \lambda < 1 \quad \Longrightarrow \quad E(\psi_\lambda) < E(\psi). \tag{6}$$

Consequently, $E|_{\mathcal{A}_m}$ has no local minimizer in the H^2 topology for any $m > 0$.

2 Proof of all-dimensional nonattainment

Proof of Theorem 1. Write

$$A(s) = \varepsilon^2 e^{-2\alpha s}, \qquad D(s) = s + A(s).$$

Since $A(s) > 0$, the function $F(s) = s/D(s)^2$ is continuous and nonnegative on $[0, \infty)$, and positive when $s > 0$. Moreover,

$$F(s) = \frac{s}{\varepsilon^4} + O(s^2) \quad (s \downarrow 0), \qquad F(s) = \frac{1 + o(1)}{s} \quad (s \to \infty). \tag{7}$$

It follows that F is bounded; set $M_F = \sup_{s \geq 0} F(s) < \infty$. For every $\psi \in H^2$,

$$0 \leq \int_{\mathbb{R}^d} F(|\psi|^2)|\Delta\psi|^2 \, \mathrm{d}x \leq M_F \|\Delta\psi\|_{L^2}^2 < \infty. \tag{8}$$

Thus the energy is finite and nonnegative in all dimensions.

Furthermore, if $E(\psi) = 0$, then $c_1 > 0$ implies $\nabla\psi = 0$ almost everywhere. Since $\mathbb{R}^d$ is connected, ψ is almost everywhere constant. The only constant in $L^2(\mathbb{R}^d)$ is zero; hence

$$E(\psi) = 0 \quad \Longleftrightarrow \quad \psi = 0. \tag{9}$$

Choose any $\psi \in \mathcal{A}_m$. For $\lambda > 0$ use the dilation (5); a change of variables gives

$$\|\psi_\lambda\|_{L^2}^2 = \|\psi\|_{L^2}^2 = m, \qquad \|\nabla\psi_\lambda\|_{L^2}^2 = \lambda^2 \|\nabla\psi\|_{L^2}^2. \tag{10}$$

Also $\Delta\psi_\lambda(x) = \lambda^{d/2+2}\Delta\psi(\lambda x)$. Writing $\rho = |\psi|^2$ and $B = |\Delta\psi|^2$, we obtain the exact scaling formula

$$E(\psi_\lambda) = c_1\lambda^2 \|\nabla\psi\|_{L^2}^2 + c_2\lambda^4 \int_{\mathbb{R}^d} F(\lambda^d \rho(y))B(y) \, \mathrm{d}y. \tag{11}$$

Using the uniform bound on F,

$$0 \leq E(\psi_\lambda) \leq c_1\lambda^2 \|\nabla\psi\|_{L^2}^2 + c_2 M_F \lambda^4 \|\Delta\psi\|_{L^2}^2 \underset{\lambda \downarrow 0}{\longrightarrow} 0. \tag{12}$$

Since $\psi_\lambda \in \mathcal{A}_m$, nonnegativity and (12) show $I_d(m) = 0$. If some $\psi_* \in \mathcal{A}_m$ attained this infimum, then $E(\psi_*) = 0$ and (9) would force $\psi_* = 0$, contradicting $\|\psi_*\|_{L^2}^2 = m > 0$. Hence the infimum is not attained. □

3 Proof of strict spreading in dimensions $1 \leq d \leq 4$

Proof of Theorem 2. With $A(s)$ and $D(s)$ as above, the derivative of F is

$$F'(s) = \frac{A(s) - s + 4\alpha s A(s)}{(s + A(s))^3}. \tag{13}$$

The combination that controls dilation is

$$\begin{aligned} 4F(s) + dsF'(s) &= F(s)\left[4 - d + \frac{2dA(s)(1 + 2\alpha s)}{s + A(s)}\right] \\ &= F(s)\left[4 - d + \frac{2d\varepsilon^2 e^{-2\alpha s}(1 + 2\alpha s)}{s + \varepsilon^2 e^{-2\alpha s}}\right]. \end{aligned} \tag{14}$$

For $1 \le d \le 4$, the bracket is strictly positive at every finite $s \ge 0$. Accordingly,

$$4F(s) + dsF'(s) \ge 0 \quad (s \ge 0), \qquad 4F(s) + dsF'(s) > 0 \quad (s > 0). \tag{15}$$

To justify differentiating the nonlinear integral in (11), note that $sF'(s)$ is bounded. Indeed, (13) makes $sF'(s)$ continuous on $[0, \infty)$ with limit zero at both $s = 0$ and $s = \infty$. Thus $4F + dsF'$ is bounded on $[0, \infty)$. For λ in any compact subinterval of $(0, \infty)$, the derivative with respect to $\log \lambda$ of the integrand $\lambda^4 F(\lambda^d \rho)B$ is bounded in absolute value by a constant times $B \in L^1(\mathbb{R}^d)$. Dominated convergence therefore yields

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\log\lambda} E(\psi_\lambda) &= 2c_1\lambda^2 \|\nabla\psi\|_{L^2}^2 \\ &\quad + c_2\lambda^4 \int_{\mathbb{R}^d} (4F(s) + dsF'(s))B(y)\,\mathrm{d}y, \qquad s = \lambda^d \rho(y). \end{aligned} \tag{16}$$

By (15), the integral is nonnegative. By (9), $\|\nabla\psi\|_{L^2}^2 > 0$ for nonzero ψ. Therefore

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}\log\lambda} E(\psi_\lambda) \ge 2c_1\lambda^2 \|\nabla\psi\|_{L^2}^2 > 0.} \tag{17}$$

Strict monotonicity and (6) follow.

For completeness, the dilation group is strongly continuous on $H^2(\mathbb{R}^d)$:

$$\lim_{\lambda \to 1} \|\psi_\lambda - \psi\|_{H^2} = 0. \tag{18}$$

To see this, first use dominated convergence for functions in $C_c^\infty(\mathbb{R}^d)$, and then pass to all of H^2 by density; the dilation operators are uniformly bounded on H^2 for λ in a neighborhood of 1, because their L^2, first-derivative, and second-derivative norms scale respectively as 1, λ, and λ^2.

Now suppose that $\psi \in \mathcal{A}_m$ were a local constrained H^2 minimizer. By (18), for $\lambda < 1$ sufficiently close to 1, ψ_λ would lie in its minimizing neighborhood. Yet (10) keeps $\psi_\lambda \in \mathcal{A}_m$ while (6) implies $E(\psi_\lambda) < E(\psi)$, a contradiction. □

4 Interpretive limits

Remark 3 (Dimensions greater than four)**.** *The proof of nonattainment uses only boundedness of F, positivity of the Dirichlet term, and mass-preserving spreading. It is therefore valid for every finite $d \ge 1$. For $d > 4$, the factor $4 - d$ in (14) can be negative. Thus the argument above does not establish strict dilation monotonicity or the absence of* local *constrained minimizers in those dimensions.*

Remark 4 (Stationary points and other models)**.** *The absence of positive-mass minimizing states is an energetic statement, not a nonexistence theorem for all stationary saddles or a dynamical scattering theorem. Likewise, excluding general constrained critical points via dilation requires justifying the dilation as a differentiable tangent variation in the selected function space. The result does not cover additional focusing terms, external confinement, prescribed boundaries, or other energy functionals with different signs.*

Source and reproducibility

A related foundational audit and scope discussion is available at https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/foundational/LOCALIZED_STATE_EXISTENCE.md.